Clinical Mortality Prediction in Heart Failure Patients: A Comparative Study of Machine Learning and Neural Networks

Tongyang Zou

April 10, 2026

Abstract

Heart failure is a leading cause of cardiovascular mortality worldwide. This study leverages a dataset of 299 clinical records to develop a predictive model for patient survival. We performed data preprocessing, feature engineering, and compared multiple algorithms including Logistic Regression, Random Forest, and Multi-Layer Perceptron (MLP). Our results show that follow-up time, serum creatinine, and ejection fraction are the most critical predictors, with the XGBoost model achieving a superior AUC compared to traditional methods.

1 Introduction

Predicting the risk of death in heart failure patients is crucial for clinical decision-making. Traditional scoring systems often fail to capture non-linear relationships between bio-markers. This paper aims to utilize the Heart Failure Clinical Records dataset to build a robust predictive pipeline.

2 Methods

This analytical pipeline follows the methodology established by Chicco and Jurman [1], prioritizing key clinical features for mortality prediction.

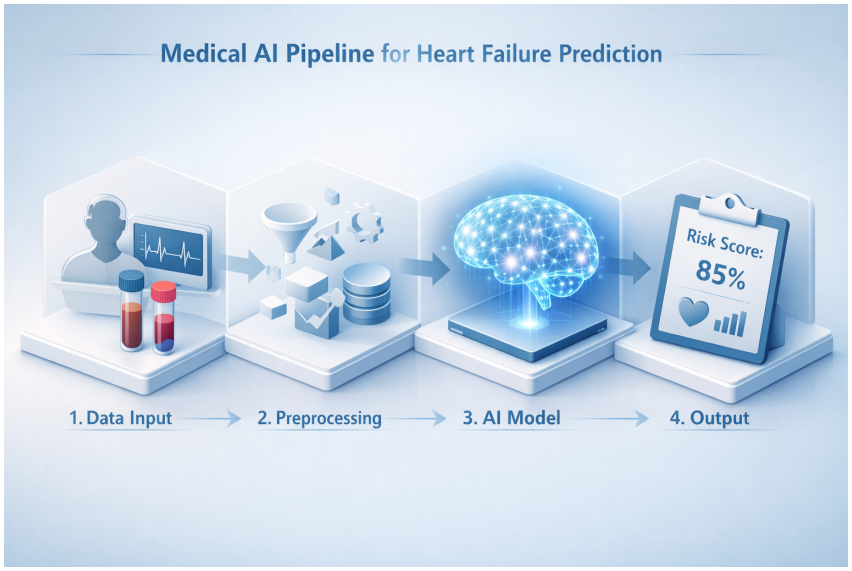

Figure 1: The proposed AI-driven clinical decision support framework for heart failure mortality prediction.

2.1 Data Preprocessing

The dataset contains 13 features. We handled outliers using clipping (1st and 99th percentiles) and applied Z-score standardization:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation. This ensures that features like *platelets* and *serum creatinine* are on the same numerical scale, allowing models like Logistic Regression and MLP to converge efficiently.

3 Results and Discussion

3.1 Exploratory Data Analysis

Before modeling, we evaluated the correlation between clinical features. As shown in Figure 2, several features exhibit significant associations with the mortality event.

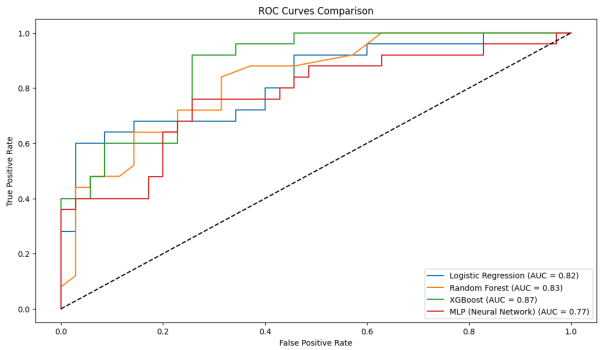

Figure 2: Correlation heatmap of clinical features and the target variable (DEATH_EVENT).

3.2 Feature Importance

Using a Random Forest classifier, we identified the top predictors. Figure 3 illustrates that *time* (follow-up period), *serum_creatinine*, and *ejection_fraction* dominate the model's decision-making process, aligning with clinical observations.

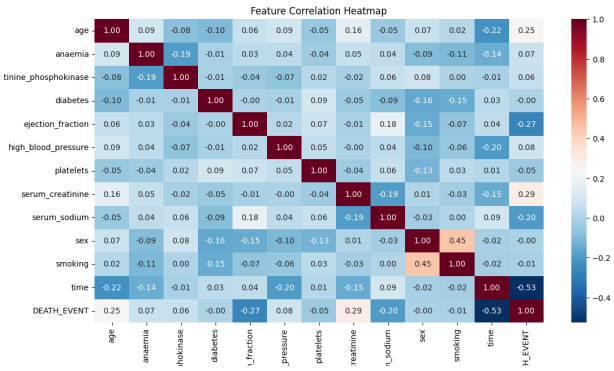

Figure 3: Feature importance ranking derived from the tree-based ensemble model.

3.3 Model Performance

We evaluated models using the F_1 score, defined as the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

The performance metrics obtained from our experimental code are summarized in Table 1.

Table 1: Comparison of machine learning model performance

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	**0.80**	**0.93**	0.56	0.82
Random Forest	0.75	0.86	0.48	0.83
XGBoost	0.78	0.83	**0.60**	**0.87**
MLP (Neural Network)	0.67	0.63	0.48	0.77

3.4 ROC Analysis

The Receiver Operating Characteristic (ROC) curves in Figure 4 demonstrate the superior discriminative power of ensemble methods (XGBoost and Random Forest) over the Multi-Layer Perceptron and baseline Logistic Regression.

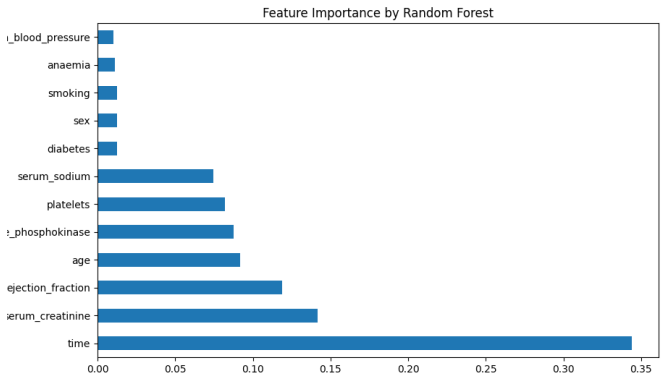

Figure 4: ROC curves comparison across four different predictive models.

4 Conclusion

This study proves that machine learning, especially gradient boosting algorithms like XGBoost, can effectively predict mortality in heart failure patients using routine clinical data. The integration of such models into clinical workflows could provide valuable early warnings for high-risk patients.

References

[1] Davide Chicco and Giuseppe Jurman. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, 20(1):1–16, 2020.